

Complex Engineering Problem Description and Overview(D1)



UNIVERSITY
OF
JOHANNESBURG

Project Topic: Wireless Network Congestion in High-Density areas

223124404

NN Sibambo

Problem Description

Wireless local area networks allow devices such as laptops and mobile phones to connect to the internet using radio signals instead of cables. In universities these networks are essential for teaching activities such as programming laboratories, software installation sessions, and online learning.

During laboratory sessions it is common for many students to download the same software at the same time. When this occurs inside a lecture hall or computer laboratory, dozens of devices begin requesting data simultaneously. Each device automatically connects to the wireless router that provides the strongest signal strength.

This connection behavior often results in an uneven distribution of users across available routers. One router may become heavily loaded while other nearby routers remain lightly loaded. The overloaded router must divide its limited bandwidth among many users. This leads to extremely slow download speeds and very long waiting times.

The engineering problem is therefore defined as the inefficient distribution of wireless network traffic among multiple routers in a high density indoor environment, resulting in severe congestion and degraded network performance.

The problem is framed as an engineering problem because it involves the performance of a wireless communication system operating under physical and network constraints. The objective is to understand why the system behaves inefficiently under heavy load conditions.

Table 1
System Boundary Definition

System Element	Included	Reason
Student laptops and smartphones	Yes	Generate network traffic
Wireless access points (routers)	Yes	Provide wireless communication
Radio frequency communication channel	Yes	Medium for signal transmission
Building environment (walls, layout)	Yes	Influences signal strength and interference

System Element	Included	Reason
External internet backbone	No	Congestion occurs locally
Remote servers hosting downloads	No	Not responsible for local traffic imbalance

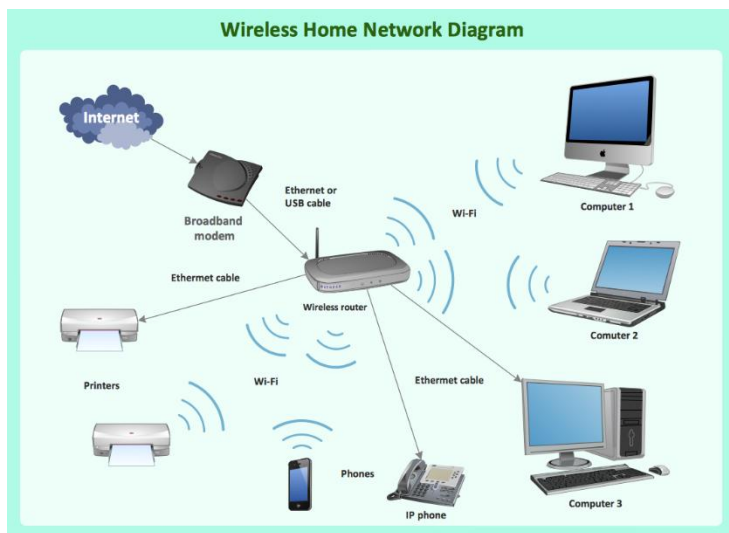
Table 2

Stakeholders Affected

Stakeholder	Impact
Students	Delayed downloads and interrupted learning
Lecturers	Reduced time for instruction
IT administrators	Increased network troubleshooting
University management	Infrastructure costs and upgrades

Figure 1

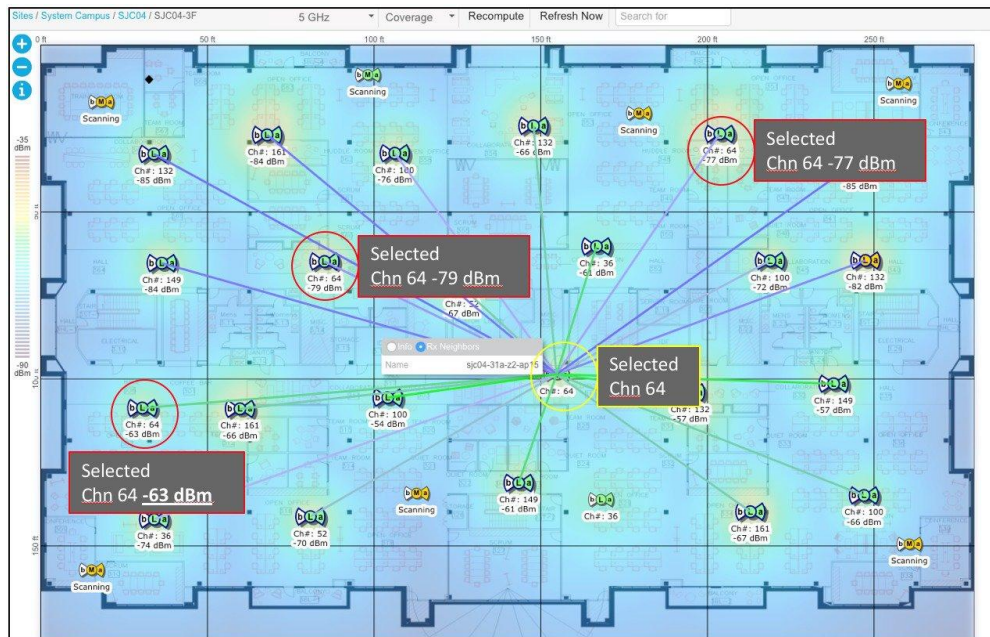
Typical Wireless Network Layout



This diagram illustrates how multiple wireless access points provide network coverage to many devices.

Figure 2

High Density Wireless Environment



This diagram

shows multiple wireless users sharing a network infrastructure.

Argument That the Problem is a Complex Engineering Problem

According to the Engineering Council of South Africa, a complex engineering problem involves several interacting systems, conflicting constraints, and requires analysis using engineering fundamentals rather than simple rules or standard procedures [3].

Systems-Level Interactions

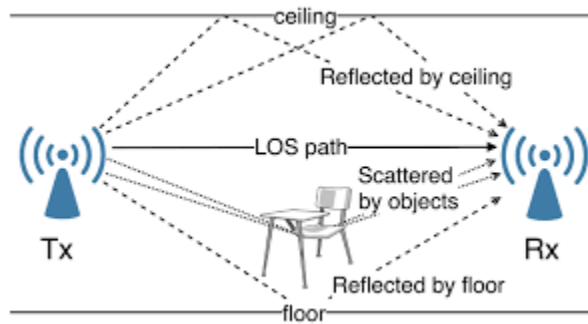
Wireless networks consist of several interacting components including user devices, access points, radio channels, and network protocols. The behavior of the entire system depends on how these components interact dynamically.

Table 3

Subsystem Interactions

Subsystem	Role
User devices	Generate network traffic
Access points	Distribute wireless bandwidth
Radio frequency channels	Carry signals between devices
Indoor environment	Causes signal attenuation and reflection

Figure 3
Wireless Signal Propagation



This diagram shows how radio signals propagate and weaken as distance increases.

Conflicting Technical Constraints

Wireless network design must balance several competing objectives.

Table 4
Conflicting Engineering Constraints

Objective	Engineering Conflict
High data throughput	May reduce fairness between users
Strong signal connection	May overload specific routers
Balanced traffic distribution	May require connection switching
Stable connections	May limit traffic redistribution

Because these objectives conflict with each other, the system cannot be optimized using simple configuration settings.

Requirement for First Principles Analysis

Analysis of this problem requires engineering laws and mathematical models.

One important principle is the Shannon–Hartley Theorem which defines the maximum theoretical data rate of a communication channel.

Shannon Capacity Equation

$$C = B \log_2(1 + \text{SNR})$$

Where

C = channel capacity in bits per second

B = channel bandwidth

SNR = signal to noise ratio

This law demonstrates that the capacity of a wireless channel is limited by bandwidth and signal quality [5].

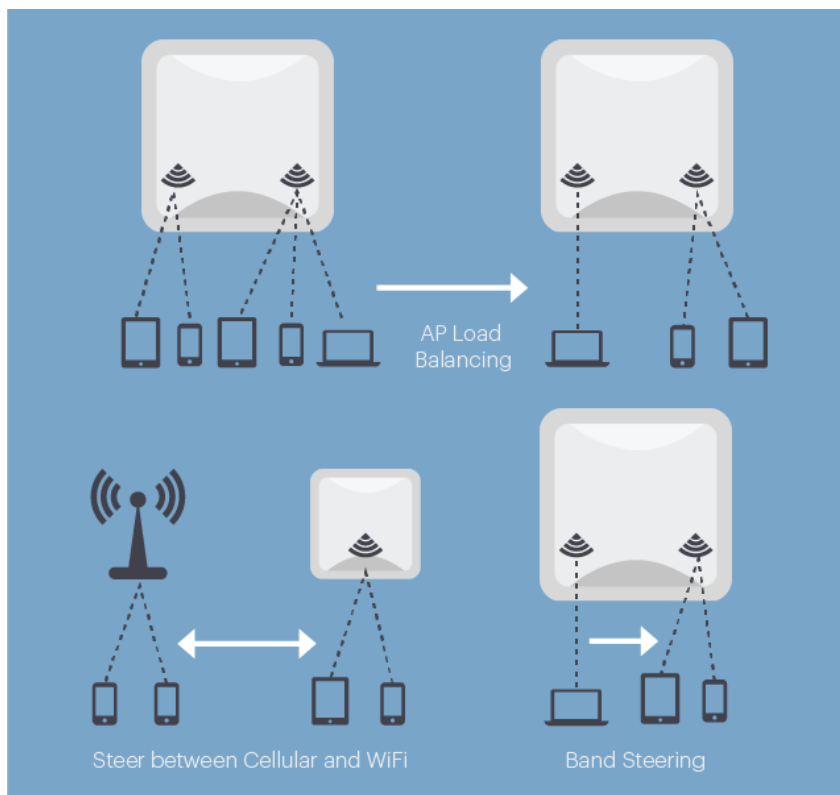
When many users share the same channel, the available capacity must be divided among them, which leads to congestion.

Absence of Standard Solutions

Standard WiFi association algorithms rely mainly on signal strength. They do not account for router load or traffic congestion. As a result, devices may connect to an already overloaded router even when nearby routers have available capacity.

Figure 4

Load Imbalance in WiFi Networks



This diagram shows how some access points become overloaded while others remain underutilized.

These characteristics demonstrate that the problem requires detailed engineering analysis and cannot be solved using simple configuration adjustments.

User Requirements

User requirements describe the expected performance of the system. These requirements do not specify design choices but instead define measurable outcomes.

Table 5
User Requirements

Requirement ID	Requirement	Measurement
UR1	Support large number of simultaneous users	Number of connected devices
UR2	Maintain acceptable download speed	Average download time
UR3	Ensure balanced distribution of users across routers	Number of users per router
UR4	Maintain stable wireless connections	Connection drop rate
UR5	Operate within existing campus infrastructure	Compatibility verification
UR6	Minimize energy consumption of user devices	Device battery usage

Table 6
Requirement Trade-Offs

Requirement	Potential Conflict
High throughput	May reduce fairness
Load balancing	May cause reconnection delays
Connection stability	May reduce traffic optimization
Low energy consumption	May limit communication overhead

Because these requirements influence each other, satisfying them simultaneously requires complex engineering analysis.

Engineering Fundamentals and Scientific Underbuild

Several engineering disciplines are required to understand the problem.

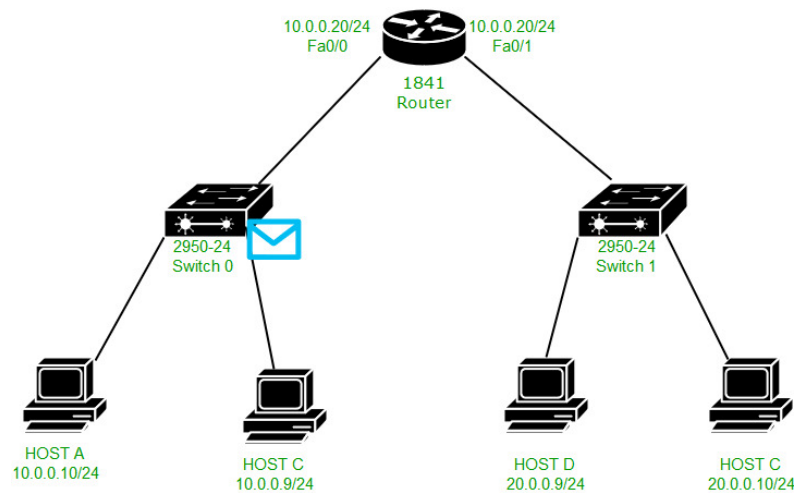
Table 7
Engineering Theory Required

Field	Application
Electromagnetic theory	Explains wireless signal propagation
Wireless communication theory	Determines data transmission limits
Information theory	Defines maximum channel capacity
Computer networking	Explains packet routing and congestion
Queueing theory	Models delays under heavy traffic
Graph theory	Represents network topology

Definition: Queueing Theory

Queueing theory studies systems where multiple users compete for limited resources. In communication networks it explains how packet delays increase as traffic demand approaches channel capacity [6].

Figure 5
Network Traffic Flow



This diagram illustrates how traffic flows through routers and communication links.

Resource Use, Environmental Impact, and Lifecycle Considerations

Engineering solutions must consider resource usage and long term sustainability.

Table 8

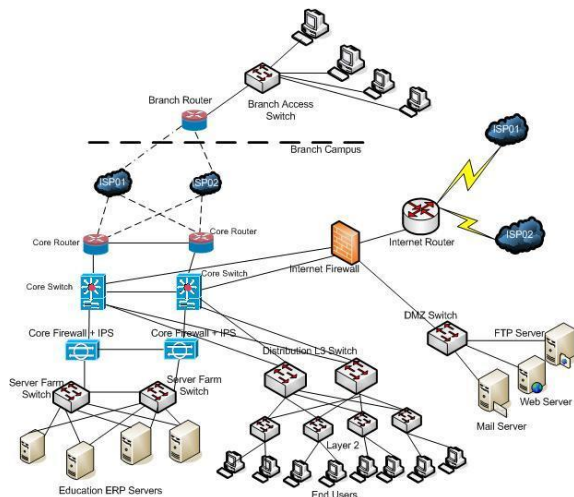
Resource and Lifecycle Constraints

Constraint	Impact
Hardware cost	Adding routers increases infrastructure expense
Energy consumption	Congestion causes repeated transmissions
Maintenance complexity	Advanced systems require skilled technicians
Scalability	Network must support future growth
Equipment lifespan	Infrastructure should operate for many years

Reducing congestion through better resource utilization may extend the lifespan of existing infrastructure and reduce energy consumption.

Figure 6

Campus Network Structure



This diagram illustrates the structure of large campus networks.

References

- [1] Wireless LAN topology diagram, Wikimedia Commons. Available:
https://upload.wikimedia.org/wikipedia/commons/6/6e/Wireless_LAN_topology.png
- [2] WLAN network architecture, Wikimedia Commons. Available:
https://upload.wikimedia.org/wikipedia/commons/8/84/WLAN_Network_Architecture.png
- [3] Engineering Council of South Africa, Accreditation Standard for Engineering Programmes.
- [4] Radio signal propagation diagram, Wikimedia Commons. Available:
https://upload.wikimedia.org/wikipedia/commons/2/21/Radio_signal_propagation.png
- [5] C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal, 1948.
- [6] L. Kleinrock, Queueing Systems Volume 1: Theory, Wiley.
- [7] Computer network topology diagram, Wikimedia Commons. Available:
https://upload.wikimedia.org/wikipedia/commons/5/5b/Computer_network_topology.png
- [8] Enterprise network architecture diagram, Wikimedia Commons. Available:
https://upload.wikimedia.org/wikipedia/commons/3/3c/Enterprise_network_architecture.png